

Solve each inequality. Show your work. Graph the solution on the number line.

1. $-5x < 108$

2. $14.7 \geq b - 12.8$



3. $w + \frac{32}{3} > \frac{17}{6}$

4. $-72 \leq \frac{y}{2}$



5. Michael takes his bike to a bike shop for repair. Labor costs \$17.50 an hour. Michael's budget allows him to spend no more than \$112 on labor.
- a. Write an inequality that can be used to determine the number of possible hours, x , the bike shop can work on Michael's bike.

Inequality	
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- b. Solve the inequality.

6. Esmeralda is saving money to take a trip to Siesta Key, Florida with her friends. Every week, she sets aside the same portion of her paycheck after taking out the \$382 she needs for her weekly expenses. After 11 weeks, Esmeralda has \$506 saved for her trip.
- a. Write an equation that can be used to determine the amount, x , in dollars that Esmeralda earns each week.

Equation	
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- b. Solve the equation.

Solve each equation. Show your work.

7. $-4(w + 7) = 108$

8. $\frac{n}{-3} + 3.8 = -7.4$

9. $3\frac{5}{6} = 8\frac{1}{5} - k$

10. $-58.2 = 6x + 9.6$

11. A server at The Main Street Grill earned \$7.80 per hour plus \$87.40 in tips.
- a. Write an equation that can be used to determine the number of hours, h , the server worked to earn \$136.15.

Equation	
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- b. Solve the equation.

12. The 9 members of a running club meet each Saturday to run 5.2 miles together. During the rest of the week, each member of the club runs the same mileage, m , on their own. The total mileage all the club members run for the week is 148.5 miles.

Complete the equation.

$$\boxed{} (m + \boxed{}) = \boxed{}$$